

First Degree Price Discrimination in the Hotelling's Linear City



Outline

- Introduction
- From Third Degree to First Degree (Perfect) Price Discrimination in a Linear City à la Hotelling
- Following-Up

Introduction

- Price discrimination and monopoly.
- Price discrimination and oligopoly:
 1. Global competition models (Cournot, Bertrand-Edgeworth);
 2. **Local-Spatial competition models** (**Line**, Circle, Spokes, Networks).
- In a linear city framework, is it possible to gradually migrate from third degree price discrimination to first degree (perfect) price discrimination?
- What happens to firms' profits under first degree price discrimination when firms are competing à la Hotelling?

From Third Degree to First Degree Price Discrimination: A Few Premises I

- A natural extension of the previous model is to have two firms, located at both ends of the Hotelling unit line, offering the same product to a unit mass of consumers and having access to information allowing them to segment the consumer space in a growing number of segments.
- Being the line of fixed length, an increased number of segments implies a decrease in each of their size. When the number of segments goes to ∞ , their length goes to 0, allowing the firms to effectively move from third degree to first degree price discrimination.



From Third Degree to First Degree Price Discrimination: A Few Premises II

- Examples of consumer's information are:
 - Gender,
 - Age,
 - Income group,
 - Geographical location ...
- We address the following questions:
 1. What are the firms' incentives to acquire customer-specific information of a given quality?
 2. How do these incentives evolve as the quality of information improves?

The Model I

- Two firms, labeled 1 and 2, are located at the two endpoints of the unit interval and they sell competing brands (at zero marginal cost).
- The utility function of a generic consumer x is defined as

$$u(x) = \begin{cases} r - p_1 - tx & \text{when buying from 1} \\ r - p_2 - t(1 - x) & \text{when buying from 2} \end{cases}$$

where $r > 0$ is the reservation utility, $p_1 > 0$ is the price charged by 1 and $t > 0$ is the transportation (or differentiation) cost.

- As usual we assume full-market coverage.

The Model II

- Both firms are able to acquire customer specific information allowing them to classify the consumers into different segments.
- It is assumed that the information partitions the unit interval into N subintervals of equal length. To simplify the analysis, it is further assumed that $N = 2^k$ with $k = 0, 1, 2, \dots$ being the measure of the quality of the information: the larger k , the finer the consumers' segmentation and the sharper the estimate of their valuation for the two brands.



- Both firms have access to information of quality k and in what follows we compare situations with (exogenously) increasing values of k .

The Model III

- Firm 1 and 2 play a sequential game with perfect information made of three stages:
 1. Given information of quality k , firms decide whether to acquire or not;
 2. Firms choose their regular prices;
 3. The firm(s) with information target(s) specific prices (discounts) to the consumer segments.
- The SPNE, or the couple of decisions on information acquisition and related prices inducing a NE in each subgame is found by means of backward induction.

Backward Induction: Step 3 and 2 - I

- Following the same framework of the previous lecture we first seek the Nash-Equilibrium prices conditional on the information acquisition decision.
- We have thus 4 subgames where equilibrium prices are sought:
 1. Neither firms acquire information (NI,NI);
 2. Both firms acquire information (I,I);
 3. One firm acquires information (I,NI) or symmetrically (NI,I).

Backward Induction: Step 3 and 2 - II

1. (NI,NI)

When both firms do not acquire information, both firms set a regular price in step 2. and step 3. is not reached. We proceed as follows:

- a. As usual we find the indifferent consumer between firms 1 and 2 by equating the utility of a generic consumer x when buying from 1 and when buying from 2

$$r - p_1 - tx = r - p_2 - t(1 - x)$$

yielding

$$\hat{x} = \frac{1}{2} - \frac{p_1 - p_2}{2t} \quad (1)$$

Backward Induction: Step 3 and 2 - III

- b. Since we assumed a fully-covered market the demand faced by firm 1 is

$$D_1 = \hat{x}$$

while the one faced by firm 2 is

$$D_2 = 1 - \hat{x}$$

- c. It follows that both firms want to solve the following profit maximization problems

$$\max_{p_1} \pi_1 = p_1 \hat{x}$$

$$\max_{p_2} \pi_2 = p_2 (1 - \hat{x})$$

Backward Induction: Step 3 and 2 - IV

which, it can be shown, have a unique solution each recoverable by their FOCs

$$p_1 = \frac{t + p_2}{2} \quad (2)$$

$$p_2 = \frac{t + p_1}{2} \quad (3)$$

- d. Therefore the Nash Equilibrium prices conditional on the information acquisition choice (NI,NI) is found solving the system of (2) and (3) resulting in

$$\bar{p}_1 = t$$

$$\bar{p}_2 = t$$

Backward Induction: Step 3 and 2 - V

implying

$$\bar{\hat{x}} = \frac{1}{2} \quad (4)$$

$$\bar{\pi}_1 = \frac{1}{2}t \quad (5)$$

$$\bar{\pi}_2 = \frac{1}{2}t \quad (6)$$

Backward Induction: Step 3 and 2 - VI

2. (I,I)

When both firms acquire information, both firms set a regular price in step 2. and their discriminatory price (promotion) in step 3. As they both know in which of the $N = 2^k$ segments each consumer is located, they are able to charge different prices for different segments. We proceed as follows:

- a. We first consider a generic segment m with $m \in \{1, 2, \dots, 2^k\}$ and we denote it as the interval

$$\left[\frac{m-1}{2^k}, \frac{m}{2^k} \right]$$

and we let p_{1m} and p_{2m} denote the prices firm 1 and firm 2 respectively charge in that segment.

Backward Induction: Step 3 and 2 - VII

- b. We then find the indifferent consumer between firms 1 and 2 in segment m by equating the utility of a generic consumer x when buying from 1 and when buying from 2 in that segment

$$r - p_{1m} - tx = r - p_{2m} - t(1 - x)$$

yielding

$$\hat{x}_m = \frac{1}{2} - \frac{p_{1m} - p_{2m}}{2t}$$

- c. Since we assumed a fully-covered market the demand faced by firm 1 in segment m is

$$D_{1m} = \hat{x}_m - \frac{m-1}{2^k}$$

while the one faced by firm 2 is

$$D_{2m} = \frac{m}{2^k} - \hat{x}_m$$

Backward Induction: Step 3 and 2 - VIII

- d. It follows that both firms want to solve the following profit maximization problems

$$\begin{aligned}\max_{p_{1m}} \pi_{1m} &= p_{1m} \left(\hat{x}_m - \frac{m-1}{2^k} \right) \\ \max_{p_{2m}} \pi_{2m} &= p_{2m} \left(\frac{m}{2^k} - \hat{x}^m \right)\end{aligned}$$

which, it can be shown, have a unique solution each recoverable by their FOCs.

Backward Induction: Step 3 and 2 - IX

- e. Therefore the Nash Equilibrium prices in segment m conditional on the information acquisition choice (I, I) is found solving the system of FOCs, resulting in

$$p_{1m}^-(k) = \frac{t(2^k - 2m + 4)}{3 \times 2^k}$$

$$p_{2m}^-(k) = \frac{t(2m + 2 - 2^k)}{3 \times 2^k}$$

implying

$$\bar{x}_m(k) = \frac{2^k + 4m - 2}{6 \times 2^k}$$

Backward Induction: Step 3 and 2 - X

- f. We then check under which conditions this indifferent consumer actually, strictly, belong to segment m (meaning that both firms sell positive quantities in m) and this is the case if and only if

$$\begin{cases} \bar{x}_m(k) > \frac{m-1}{2^k} & \iff m < 2^{k-1} + 2 \equiv m_+(k) \\ \bar{x}_m(k) < \frac{m}{2^k} & \iff m > 2^{k-1} - 1 \equiv m_-(k) \end{cases}$$

We note that such conditions are always met for $k = 0$ or $k = 1$, while for $k \geq 2$ the firms compete only in

$$m_-(k)$$

and

$$m_+(k)$$

Backward Induction: Step 3 and 2 - XI

- g. Therefore, in segments 1 to $m_-(k)$ firm 1 acts as a monopolist charging

$$p_{1-}(k) = \frac{t(2^k - 2m)}{2^k}$$

as the unique Nash Equilibrium price in segments from 1 to $m_-(k)$ conditional on the information acquisition choice (I,I).¹

- h. Similarly, in segments $m_+(k)$ to 2^k is firm 2 acting as a monopolist, charging

$$p_{2+}(k) = \frac{t(2m - 2^k - 2)}{2^k}$$

as the unique Nash Equilibrium price in segments $m_+(k)$ to 2^k conditional on the information acquisition choice (I,I).²

Backward Induction: Step 3 and 2 - XII

- i. Above points yield the Nash Equilibrium profits conditional on the information acquisition decision (I,I)

$$\pi_1(\bar{k}) = \pi_{1-}(\bar{k}) + \pi_{1m}(\bar{k})$$

$$\pi_2(\bar{k}) = \pi_{2m}(\bar{k}) + \pi_{2+}(\bar{k})$$

It is worth to note that both functions exhibit a U-shape determined by the interplay of two conflicting forces: higher competition and surplus extraction. The first effect dominates when the quality of information k is low (hence with third degree price discrimination), determining lower prices, while the second effect becomes predominant when k is high (hence with almost first degree price discrimination), determining a firm's increased ability of raising prices extracting more surplus.

Backward Induction: Step 3 and 2 - XIII

- j. Finally, it can be shown³ that when $k \rightarrow \infty$, hence with (almost) first degree price discrimination,

$$\bar{\pi}_1 = \frac{t}{4} \quad (7)$$

and

$$\bar{\pi}_2 = \frac{t}{4} \quad (8)$$

Backward Induction: Step 3 and 2 - XIV

3. (I,NI)

When only one firm acquires information, say 1, both firms set a regular price in step 2. and the discriminatory price (promotion) in step 3. is set by firm 1 only. As firm 2 lacks the necessary information to treat the various segment independently, it is no longer possible to solve for the equilibrium in each segment separately and the problem becomes significantly more complex to be solved. However, the solution might be characterized as follows:

- a. The set of segments can again be divided into three groups. Going from left to right, we have that firm 1 acts as a constrained monopolist in the first group of segments; then comes a group of segments where both firms have positive demand; finally, in the last group of segments, it is firm 2 that acts as a constrained monopolist.

Backward Induction: Step 3 and 2 - XV

- b. We note then that 1, the informed firm, is a constrained monopolist on a larger number of segments than 2, the uninformed firm. We further note that competition takes place in segments located in the second half of the interval, meaning that it is only 1 that is able to "steal" some 'loyal' consumers from 2. For instance, for $k = 3$ (8 segments), firm 1 attracts all consumers in segments 1 to 5, some consumers in segments 6 and 7 and leaves all consumers in segment 8 to firm 2.

Backward Induction: Step 3 and 2 - XVI

- c. It is worth noting that, it can be shown, that the uninformed firm's profits monotonically decrease with the quality of information, while the informed firm's profits are non-monotonic in k as the same two forces of higher competition and surplus extraction are at play. When the quality of information is low ($k = 1, 2$), profits fall below the non-discrimination case because of the other firm's aggressive reaction: knowing that the rival has committed credibly to price discriminate, the uninformed firm lowers its regular price significantly. While if the information becomes more precise ($k > 3$), the informed firm is able to extract sufficiently more surplus, so as to take advantage of the information benefits in spite of the uninformed firm's defensive reaction.

Backward Induction: Step 3 and 2 - XVII

- d. Eventually, it can be shown⁴ that the Nash Equilibrium prices conditional on the information acquisition decision (I,NI) are

$$\bar{p}_1 = \max(p_2 + t(1 - x), tx)$$

and

$$\bar{p}_2 = \frac{t}{2}$$

- e. Finally, it can be shown⁵ that when $k \rightarrow \infty$, hence again with (almost) first degree price discrimination,

$$\bar{\pi}_1 = \frac{9t}{16} \tag{9}$$

and

$$\bar{\pi}_2 = \frac{t}{8} \tag{10}$$

¹ See (Belleflamme 2010, 204)

² *Ibidem*

³ *Ibidem*

⁴ See (Belleflamme 2010, 205)

⁵ See (Belleflamme 2010, 206)

Backward Induction: Step 1 - I

- In step 3. and 2. we found the equilibrium prices conditional on the information acquisition decisions, namely the equilibrium prices conditional on (NI,NI), (I,I) and (I,NI).
- Now we seek the equilibrium acquisition decisions as follows:
 1. Equations (5) to (10) give us the equilibrium profits conditional on each possible couple of information acquisition decisions.
 2. We are therefore able to construct a matrix of payoffs to easily detect Nash-Equilibria in ,

	NI	I
NI	$t/2, t/2$	$t/8, 9t/16$
I	$9t/16, t/8$	$t/4, t/4$

3. Then the NE is (NI,NI) if $k \leq 2$ while (I,I) if $k \geq 3$.

Backward Induction: Step 1 - II

- It is worth noting that the second NE is determined by a prisoner's dilemma type of situation, both firms would indeed be better off by choosing not to acquire information, still they opt to do so reducing their payoffs.
- Such NEs, and their linked SPNEs, are due to the two competing effects that information acquisition generates: a *surplus extraction effect* and a *competition effect*. Namely, when the quality of information is low, the competition effect dominates and firms prefer to credibly commit to set a uniform price, by not acquiring the information. When the quality of information improves, the surplus extraction effect becomes stronger and it is a dominant strategy for the firms to acquire the information and price discriminate.

Summarizing

- In a competitive setting, customer-specific information impacts firms in two conflicting ways. On the one hand, it allows firms to extract more surplus from each consumer. On the other hand, it exacerbates price competition. When the quality of information is sufficiently large, the former effect dominates the latter. Then, firms use the information and price discriminate at equilibrium. However, they may well be better off if they could jointly agree not to use the information.
- What about consumer's welfare? It can be shown that price discrimination raises consumer's welfare, still asking for public policy intervention is not an easy call, for firms' profits are bound to decrease.

Following-Up

- What about price discrimination in more sophisticated local-spatial competition models (Salop, Spokes, Networks)?
- What about price discrimination in two-sided markets?

References

Belleflamme, Paul and Martin Peitz. 2010. *Industrial Organization: Markets and Strategies*. New York: Cambridge University Press.